

Tutorial Sheet – Data Warehouse and Data Mining

Q1(a) Explain technical metadata and business metadata in a data warehouse. Illustrate with a hospital management system example.

Q1(b) Describe the Extract–Transform–Load (ETL) process. Explain one transformation related to data integration and one related to data standardization.

Q2(a) Explain the two-tier data warehouse architecture. Compare it with three-tier architecture in terms of scalability and performance.

Q2(b) What is data preprocessing? Identify any two data inconsistencies and explain how they are handled.

Q3(a) A telecom company wants to analyze call duration and revenue across customer type, location, and time. Identify fact table, measures, and dimensions.

Q3(b) Design a STAR schema for the above scenario and justify its suitability.

Q4(a) Propose a Snowflake schema for analyzing student enrollments across college, department, program, and academic year.

Q4(b) Give two reasons why Snowflake schema is preferred over Star schema in this case.

Q5(a) Recommend a suitable OLAP model for fast query response and moderate data size. Justify.

Q5(b) Differentiate between HOLAP and ROLAP based on storage and performance.

Q6 Numerical:

A sales data warehouse contains Month, City, Category dimensions and Revenue measure.

(a) Perform Roll-Up to find yearly revenue per city.

(b) Apply Dice for City=Delhi and Category=Electronics.

Q7(a) Why is time dimension mandatory in most data warehouses?

Q7(b) Mention two fact table design mistakes and their impact.

Q8 Short Notes:

- Surrogate key vs Natural key
- Data granularity
- OLTP vs Data Warehouse
- Role of aggregation in OLAP
- Importance of data cleaning